UFSM00066 · Sistemas Dinâmicos
Lista de exercícios 5
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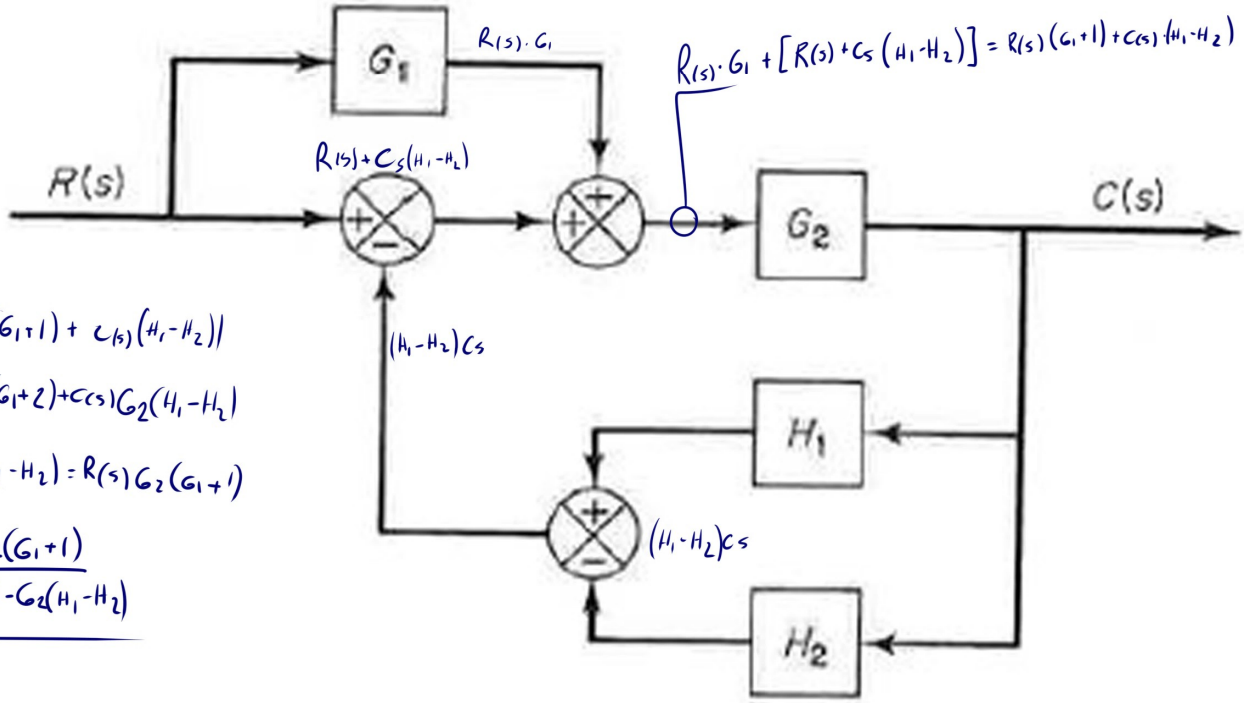
1c



$C_s = G_2(R(s)(G_1+1) + C(s)(H_1-H_2))$

$C(s) = R(s)G_2(G_1+2) + C(s)G_2(H_1-H_2)$

$C(s)[1 - G_2(H_1-H_2)] = R(s)G_2(G_1+1)$

$$\frac{C(s)}{R(s)} = \frac{G_2(G_1+1)}{1-G_2(H_1-H_2)}$$

2c

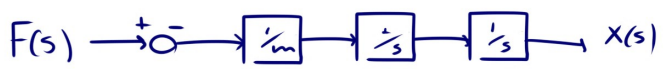
$$G'' = \frac{G_1 G_2(G_3+G_4)}{1+G_1G_2H} \qquad G' = \frac{G_1\cdot G_2}{1+G_1G_2H} \qquad \frac{Y(s)}{U(s)} = \frac{G}{1+G} = \frac{G_1G_2(G_3+G_4)}{1+G_1G_2H+G_1G_2(G_3+G_4)}$$

3 $M_x''(t) = f(t) - Bx'(t) - Kx(t) - K_1x(t)$

$M_x''(t) + Bx'(t) + (K+K_1)x(t) = f(t)$

$Ms^2X(s) + BsX(s) + (K+K_1)X(s) = F(s)$

$s^2X(s) = \frac{1}{m}(F(s) - BsX(s) - (K+K_1)X(s))$



4 $K_1 = \frac{1}{K_2}$ $\qquad Y = \frac{GK_2}{L+GK_2}R + \frac{G}{1+GK_2}$

$Y(s) = \frac{GK_2}{L\cdot GK_2}R(s) + \frac{G_1(L-K_1K_2)}{L+GK_2} = D(s)$